

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457292.7015 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.027
Transit depth (Rp/Rs)^2	1.36 +/- 0.63 [%]
Orbital Inclination (inc)	89.09 +/- 2.18
Airmass coefficient 1 (a1)	1.035 +/- 0.011
Airmass coefficient 2 (a2)	-0.0307 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#1 - [327, 231]
Optimal Aperture	3.94
Optimal Annulus	10.69
Transit Duration (day)	0.1293 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.116 $\pm$ 0.027 vs 0.10538 (deviation 0.34 σ)
Duration fit vs published	0.1293 d vs 0.1167 d (deviation 2.48 σ)
Mid-time O-C	-5.2 min
Depth SNR	3.7
Residual scatter	1.07 %

Light curve

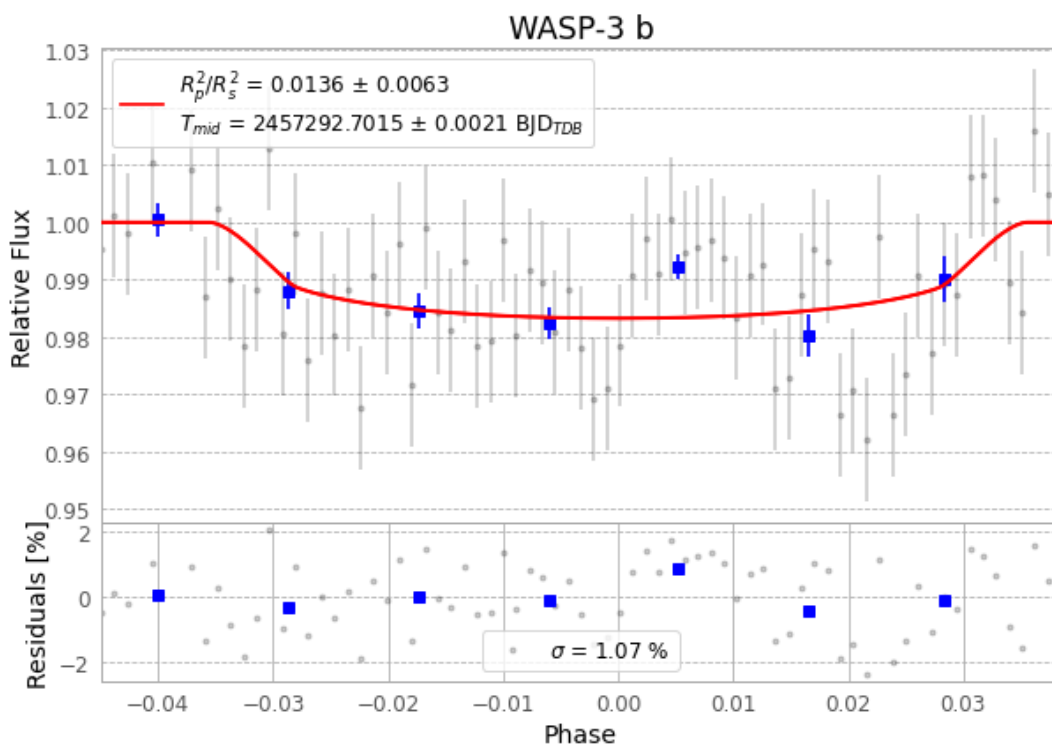


Figure 1 — FinalLightCurve_WASP-3 b_2015-09-26.png (SHA-256 65657ae093328478...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [253, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8b1edf37d6cfb20a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-3150927024812.FITS → WASP-3150927063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150927.log (245f2467a6fc...), FinalLightCurve_WASP-3 b_2015-09-26.png (65657ae09332...), FinalLightCurve_WASP-3 b_2015-09-26.csv (2cc7d858aa84...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:09Z.